3.2.2 The immune system

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	primary defences and non-specific defences against pathogens	Primary defences to include mucus and cilia in the respiratory tract, lysozyme in tears and stomach acid **AND** non-specific immune responses to include phagocytosis and inflammation.
(b)	the mode of action of phagocytes	To include the roles of cytokines, opsonins, phagosomes and lysosomes.
(c)	the different roles and modes of action of B and T lymphocytes in the specific immune response	To include clonal selection and clonal expansion, plasma cells, T helper cells, T killer cells and T regulatory cells.
(d)	the secondary immune response and the role of memory cells in long term immunity	To include T memory cells and B memory cells.
(e)	the structure and general function(s) of antibodies	To include descriptions of antibody structure from diagrams **AND** an outline of the action of opsonins and agglutinins.
(f)	how individuals can be tested for TB and HIV infection	To include antibody tests and antigen tests for both diseases **AND** the Mantoux test for TB.
(g)	the differences between active and passive immunity, and between natural and artificial immunity	To include examples of each type of immunity.
(h)	how allergies can result from hypersensitivity of the immune system.	To include an outline of the sequence of events in a typical allergic response to allergens such as pollen (hay fever).